

Our answers to Prateek's questions, and where Protopus is going

Soham, founder · 2 October 2026 · all figures in ₹ at ₹88 per US\$ unless stated

What changed after your feedback

- **We now pitch Proto, the AI engineer.** [Baker](#), Mixer and the [Probe Matrix](#) are how Proto reaches the physical world. They are not the story on their own.
- **Prototyping, not production.** [Baker](#) covers the first 1–10 boards of a design. Volume still goes to a fab, with files that already worked on a real board.
- **Softer claims, all marked as projected:** lower cost per try, shorter lead time, fewer skills needed. We no longer say “replaces a fabrication order” or “send at night, pick up in the morning”.
- **Baker first.** The ask is ₹15,00,000 for nine months, or ₹10,00,000 lean, and it funds [Baker](#) and Proto only. Mixer and the [Probe Matrix](#) are step 2, funded after the proof of concept.
- **Payback for the customer:** [Baker](#) at ₹2,99,000 pays for itself after 47 boards compared with ordering from JLCPCB, which is about 6 months for a startup and 12 months for a college lab.
- **Scale:** one [Baker](#) makes 750–1,250 boards a year. At 10,000 boards or more, a fab is the right tool and we say so.

Updated deck: PROTOPUS_Pitch_Deck_Baker_First (20 slides). The numbers come from PROTOPUS_Economics_Model.xlsx and PROTOPUS_Baker_Mixer_BOM_and_Costing.xlsx. Prices are projected. Cycle times are engineering targets, and the proof of concept will measure them.

1 · The moat is Proto. Why it still needs machines

“The strongest moat is hidden in the idea... pitch Proto instead of two hardware robots. Don’t project this as a desktop PCB mill, which has a rough history.”

Agreed: the company is Proto. Reasoning models can already run software end to end. A commerce graduate can “vibe-code” an app, because the model reaches every tool it needs through APIs and connectors. Hardware has no API to the physical world. A model can draw a schematic (Flux.ai already produces Gerbers), and Claude can already drive Fusion 360 and Blender through connectors. But no model can etch copper, place a part, or tell whether the I²C bus on a real board is alive. **To build hardware, an AI needs machines it can drive, and it needs instruments it can read.**

That is the gap the machines fill. [Baker](#) is Proto’s hands: it makes the board. Mixer is Proto’s eyes. Its signal generator and analysers let Proto apply real signals, reproduce sensor faults and read the response. The [Probe Matrix](#) accessory puts spring probes under every net, so Proto can test and program the board without a human moving probes around. The machines are not the moat. They are what makes the moat possible: every run logs design → machine settings → measurements → outcome. A tool that only produces files never sees the real board, so it can never collect this data.

The expertise curve: where people drop out of hardware

People learn practical skills by doing: try, look, adjust (Kolb’s experiential learning cycle). The speed of that loop sets the speed of learning. Hardware makes a beginner climb eight rungs, and each rung loses people:

Rung	You first need to know	Typical bottleneck	Who carries it with Protopus
1 · Idea → spec	How to state requirements	Vague goals, scope creep	Proto interviews you and writes the spec
2 · Parts	Datasheets, footprints, stock	Hours of searching; parts go out of stock	Proto picks in-stock parts through parts APIs
3 · Schematic	Circuit theory, reference designs	Errors that only show on the bench	Proto starts from reference designs, runs ERC (Flux)
4 · Layout	An EDA tool, DRC, manufacturability	Weeks of learning a tool	Proto drives the EDA tool; Baker -specific DRC
5 · Fabrication	Gerbers, vendors, customs	7-20 days to India per attempt	Baker makes the board the same day (target)
6 · Assembly	SMD soldering, stencils, reflow	Skill and equipment	Baker cuts the stencil, pastes, reflows; a printed jig guides placement
7 · Bring-up	Scope, logic analyser, firmware	You can't tell what's wrong	Bench instruments now; Mixer + Probe Matrix in step 2. Proto explains in plain words
8 · Iterate	Root-cause debugging, DFM	Slow loops mean fewer attempts	Proto proposes the fix, then hands off to a fab for volume

Rungs 1–4 are already being solved in software. Rungs 5–8 are physical, which is why the AI-only players stop at the file. Each rung we remove brings in a new group of builders: students, designers, domain experts and first-time founders. Lovable reached \$100M ARR in 8 months by removing the coding rung for people who can't code. **Ease is the market.**

Not a desktop PCB mill. A mill automates one step (rung 5, partly) for people who are already experts, and that category has a rough history: Bantam Tools stopped selling its PCB mill in 2025. [Baker](#) runs 15 steps in one coordinate frame, from bare FR4 to a reflowed board. Proto plans each step, so the user does not need to be an expert.

2 · Softer claims, marked as projected

“Avoid strong claims like “replaces a fabrication order”. An investor in hardware manufacturing... how would he take it?”

Fair point. We never meant to replace fab houses. Protopus fills the gap between a finished design and a board you can test in the real world. Think of a plastic bottle: you 3D-print it to check shape, feel and fit, and you injection-mould the first thousand. Plastic has had a desktop prototyping path for years. Electronics has never had a desktop path that ends in a finished, soldered and tested board. [Baker](#) is that path. Once the design is frozen, Proto sends the fab files that have already worked on a real board, so a fab gets a cleaner first order and is not losing one. **For an investor in manufacturing, we sit upstream of the fabs and feed them.**

What we said before	What we say now	Basis
“Replaces a fabrication order”	For the first 1–10 boards of a design: a lower cost per try and a same-day lead time	₹334 in consumables vs ₹6,721 for the same board from JLCPCB landed in India (7–20 days)
“Send the design at night, pick it up in the morning”	About 2 hours per board, 15–20 minutes of it hands-on (target)	Cycle-time model, 81–141 minutes; measured in the PoC
“Fully automatic”	Steps 1–12 automatic; you drop parts into a jig and start the reflow	See question 5
“Eight instruments replace your bench”	Analog Discovery 3-class test, run by Proto (step 2)	Mixer tech stack, ₹13,740 production cost (projected)

Every motor speed, processing time and jig-print time in the deck is a projected engineering estimate. The proof of concept exists to measure them, and the deck labels them “target” or “projected”.

3 · Baker first: one product at a time

“Two products for a first raise of ₹10L is a bit aggressive. Step by step: maybe start with Baker first?”

Agreed, and the deck now says so. The two machines are the **vision**: one makes the hardware, and the other tests it, programs it and feeds back to Proto. The **raise** funds one of them. Step 1 is Proto + Baker. During step 1, Baker's boards are tested on ordinary bench instruments. Mixer and the Probe Matrix are step 2, funded by the seed round after the proof of concept.

Step	What	Funded by	Proof point
1 · now	Proto + <u>Baker</u>	₹15L (lean: ₹10L)	All 15 <u>Baker</u> steps run; 20 proof-of-concept boards; Proto goes from brief → Gerbers → <u>Baker</u> job; cycle time measured
2 · after the PoC	Mixer + <u>Probe Matrix</u> ; <u>Baker</u> pilots via China	Seed, ~₹1.5 Cr (indicative)	20 Bakers + 100 Mixers built turnkey in China; tested and shipped from India
3 · ecosystem	Proto runs it all, with any 3D printer	Revenue + Series A	One plain-English brief in, one tested prototype out

₹15,00,000 · 9 months (recommended)	₹	Lean ₹10,00,000 · 6 months	₹
<u>Baker</u> prototype build (70-line BOM, duty, 15% contingency)	3,73,418	<u>Baker</u> prototype build	3,73,418
<u>Baker</u> Rev B + ~150 test boards of consumables	1,60,000	<u>Baker</u> Rev B	1,00,000
One software + firmware engineer, 9 × ₹50,000	4,50,000	Engineer, 6 months	3,00,000
Proto software + compute, 9 × ₹15,000	1,35,000	Proto compute, 6 months	90,000
Testing, tooling, lab safety	1,50,000	Testing + tooling	1,00,000
IP + legal, outreach, reserve	2,31,582	Reserve (no lab set-up, IP or outreach)	36,582
Total (no founder salary)	15,00,000	Total; fundraising starts in month 4	10,00,000

4 · Market and competition: why we are different

“A thorough competition analysis and market study is needed... look at the competition and their costing.”

The market in one paragraph. Protopus sits where two large markets meet: printed circuit boards (₹6.4 lakh crore worldwide; ₹55,000 crore in India, growing about 15% a year) and test and measurement (₹3.1 lakh crore worldwide; ₹8,300 crore in India). Our beachhead is far smaller and countable. India has 5,868 AICTE engineering colleges and about 2,250 hardware startups (1% of the 2.25 lakh DPIIT-recognised startups). That is 8,118 buyers, or **₹243 crore** of Baker sales at one machine each (₹271 crore with Mixer), plus ₹8.1 crore a year in Proto plans. Behind them come 10,000 Atal Tinkering Labs, each with a ₹20 lakh grant over five years, and 50,000 more announced. Around the world there are about 2,500 Fab Labs, 33 million active Arduino users and 2.3 million Hackster members. **Every competitor solves one step.** Mills and printers make a bare board for an expert. Pick-and-place machines only place parts. USB instruments measure, but a human has to read the screen. Fabs deliver in weeks, and AI design tools stop at the file. **Protopus is the only offer that runs the whole loop, and the only one where an AI drives the machines.** That loop is what makes prototyping easy enough for a beginner, and every board it runs teaches Proto something.

Competitor pricing, and where each one loses to us

Product	What it does	List price	Where it loses to Protopus
LPKF ProtoMat (S-series)	Pro in-house PCB mill	₹8.6–24.5 lakh per machine; ~₹84 lakh for a full S104 line (2019)	Separate machines for mask, plating and assembly; built for experts; no test, no AI
Voltera V-One	Prints conductive ink; dispenses paste; reflow	~₹3.1 lakh (\$3,500)	Ink traces, not etched FR4 copper; no solder mask, no laser stencil, no test, no AI
Makera Carvera / Carvera Air	Desktop CNC mill	₹4.8 lakh / ₹2.2 lakh	Mills only: no mask, stencil, paste, reflow or test

Product	What it does	List price	Where it loses to Protopus
Bantam Tools PCB mill	Desktop PCB mill	Discontinued 2025	The “rough history”: one step, for experts
Neoden YY1	Pick-and-place	~₹2.6 lakh	Places parts only; no board, no test
Digilent Analog Discovery 3	USB oscilloscope + logic + supplies	~₹34,000	A human reads the screen; it doesn't know the design
Moku:Go / Red Pitaya / ADALM2000	USB or FPGA instruments	₹21,000–70,000	Same: measurement without the design or a verdict
JLCPCB / PCBWay	Fab + assembly	₹6,721 per assembled try, landed	7–20 days door to door; minimum orders; customs paperwork
Indian 24-hour fab	Bare boards, fast	₹3,000–8,000 for 5	Bare board only: stencil, assembly and test are extra
Flux.ai	AI PCB design (SaaS)	₹1,300–4,300 per editor per month	Stops at the file. Proto uses its API and goes on to the real board
Protopus: Proto + Baker	Board, mask, stencil, paste, reflow; AI-planned	₹2,99,000 + ₹9,999/yr Proto (projected)	Mixer ₹34,999 and the Probe Matrix are added in step 2

What the same loop costs with competitor parts: a Carvera mill (₹4.84 lakh), a Neoden YY1 placer (₹2.64 lakh) and an Analog Discovery 3 (₹34,232) come to **₹7.82 lakh before import duty**. That set-up still has no solder mask, no stencil and no AI tying it together. [Baker](#) + Mixer is ₹3.34 lakh (projected list). Our production cost is ₹1,44,842 for [Baker](#) and ₹13,740 for Mixer, which gives 51.6% and 60.7% gross margin on build cost. After all costs, the margin is about 33% and 36%.

List prices are converted at ₹88 per US\$ and exclude Indian duty and GST. Sources (one per row) are in the Competitors sheet of the economics model: LDKF via Design World and exapro, Voltera, Makera, ToolGuyd, EEVblog, RS/RobotShop, Liquid Instruments, Red Pitaya, Mouser, JLCPCB, zbotic.in and flux.ai. Market figures: IMARC, Grand View, AICTE, DPIIT/TiE, Atal Innovation Mission, Union Budget 2025-26, Fab Foundation, Arduino and Hackster.

5 · Automation, stated honestly

“Send the design at night, pick it up in the morning is too aggressive: we have to load boards, place components... it's not fully automated.”

Correct: [Baker](#) is not 100% automatic, and we no longer claim it is. Here is exactly what is automatic and what is not. **Boards don't need loading every time.** A refill magazine feeds the next FR4 blank, and the operator only refills the magazine. The **placement jig is generated automatically from the Gerber** and printed in parallel with the board. The user drops the parts into the jig and puts the jig on the reflow plate. Reflow and jig-making happen inside the machine but are not hands-free: the user moves the jig in and out.

Stage	Automatic?	Minutes (target)	What the user does
Steps 1–2 · feed the blank from the refill magazine, probe the height map	Yes	5	Refill the magazine now and then
Step 3 · isolation mill + drill	Yes	30–55	Nothing
Steps 4–9 · clean, coat the mask, UV image, develop, cure	Yes	16–27	Nothing
Steps 10–12 · laser-cut the stencil, clamp, paste, release	Yes	8–13	Nothing
Placement jig · generated from the Gerber, printed in parallel	Yes (parallel)	0 extra	Nothing
Step 13 · place ~40 parts into the jig	No	7–14	Drop the parts in with a vacuum pen
Step 14 · low-temperature reflow, 138 °C	Profile is automatic	10–12	Put the filled jig on the plate
Step 15 · test	Proto runs the plan	5–15	Connect the bench now; with the Probe Matrix in step 2, just drop the board in
Total per board		81–141	≈15–20 min hands-on; 3–5 boards per 8-hour shift

The new promise: “Start a board after lunch and test it the same afternoon”: about 2 hours per board, all of it projected until the proof of concept measures it.

6 · What it costs, and how many boards pay it back

“How much is this machine going to cost? Say X lacs: how many boards are created, and does it cancel my cost?”

₹2,99,000 Baker list price (projected, ex-GST)	₹334 Baker consumables per board	₹6,721 same board, JLCPCB landed in India	47 boards to pay back Baker vs JLCPCB		
Customer buys	Compared with	Saving per board	Boards to pay back	Startup, 2 boards/wk	Lab, 1 board/wk
Baker · ₹2,99,000	JLCPCB to India	₹6,387	47	5.9 months	11.8 months
Baker · ₹2,99,000	Indian 24-h fab	₹5,166	58	7.2 months	14.5 months
Baker + Mixer (step 2) · ₹3,33,999	JLCPCB to India	₹6,387	53	6.6 months	13.2 months

Adding the first year of the Proto plan (₹9,999) moves [Baker](#)’s payback to 49 boards. Our count is deliberately conservative: it leaves out the engineer’s time and the 7–20 days spent waiting, which is usually worth more than the boards. The comparison with the Indian fab is conservative too, because that price buys bare boards only. If the price moves, the payback moves with it (bundle with Mixer, against JLCPCB):

Baker price	₹2,99,000	₹3,49,000	₹3,99,000	₹4,49,000
Boards to pay back (Baker only)	47	55	63	71
Boards to pay back (with Mixer)	53	61	68	76
Our gross margin on Baker	51.6%	58.5%	63.7%	67.7%

From our side: each [Baker](#) earns ₹1,54,158 of gross profit (about ₹99,000 after all costs), so 10 Bakers sold return the whole ₹15 lakh round on gross profit, and 16 do so on all-in profit.

7 · Scale: 1,000? 10,000? 1 lakh boards?

“Innovation is good but may not be scalable in a production environment.”

[Baker](#) is not built for production volume, and we don’t claim it is. The scaling question has two answers: the **customer’s boards** and **our machines**.

Boards a year	Bakers needed (1 shift)	Right tool	Why
10	1	Baker	Same-day iterations beat fab minimums and shipping
100	1	Baker	One machine covers a busy lab or startup
1,000	1–2	Baker + a fab	Iterations on Baker ; any batch of 10+ identical boards goes to a fab
10,000	14	Fab / EMS	Production. Proto exports the checked Gerber/BOM/placement files
1,00,000	134	Fab / EMS	Not what Baker is for, and we don’t claim it

What scales is the number of people who can prototype, not the boards per machine. A college lab with one [Baker](#) runs 750–1,250 boards a year (3–5 per shift, target). Proto is software, so each extra user costs only compute. **Our own manufacturing scales on a turnkey model:** parts sourced and assembled in China, then factory-tested, with final test, minimal assembly and shipping from India. Production cost is ₹1,44,842 per [Baker](#) at 100 units, and RFQs are planned from 5 to 5,000 units. We sell in India first, then globally. The China master plan sets out the plant, the suppliers and the quality gates.

8 · The vision: one brief in, one tested prototype out

The ecosystem. Proto runs [Baker](#) (makes the board), Mixer with the [Probe Matrix](#) (tests and programs it), and **any 3D printer the user already owns**. Protopus will never sell a printer.

- **Proto runs on large reasoning models.** They break a goal into steps and call tools: EDA through the Flux.ai API, CAD through Fusion 360 and Blender connectors, parts suppliers and our machines. Every project lives in its own **repository**, holding the spec, schematic, Gerbers, firmware, enclosure, jig, test plans and measured results. Each revision can be compared with the last, like code.
- **Electronics:** Proto produces a BOM of in-stock parts, each with a purchase link. Affiliate links are a possible extra revenue line.
- **Mechanical:** Proto designs a 3D-printable enclosure sized from the finished board, measured inside the same environment: component heights from the jig, outline and holes from the Gerber.

Example: a smartwatch with 10 physical buttons

#	What happens
1	You describe it in plain English: “a smartwatch with 10 physical buttons, heart-rate sensor, a week of battery”.
2	Proto checks feasibility and limits (size, battery, button count against wrist space) and proposes the best prototype for the goal.
3	Proto writes the requirements back in plain English for you to approve.
4	Proto designs the electronics: MCU, sensors, buttons and power. It produces the PCB and a BOM with purchase links.
5	Baker makes the board. Your 3D printer prints the case, sized from the measured board.
6	Proto writes and flashes the firmware through Mixer and the Probe Matrix .
7	Mixer tests every function with real signals, flags failing parts or nets, and gives a plan of action.
8	You wear it and tell Proto what felt off (“buttons too stiff, screen too dim”).
9	Proto plans the next revision, and the loop runs again. The final files go to a fab when you are ready for volume.

Protopus is a company that builds smart prototyping solutions: smart machines and accessories, run by an AI engineer. [Baker](#) is the first product, and Mixer and the [Probe Matrix](#) come next. Software and hardware teams keep improving both. Manufacturing is turnkey in China, with testing, minimal assembly and shipping from India, to India first and then the world. **The one-stop solution for prototyping: idea to innovation, without the friction.**

Links: <https://protopus-kappa.vercel.app/#/baker> · <https://probe-matrix.vercel.app/>